SOHAN GADEWAR

Pune, Maharashtra, India

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Education

Vishwakarma Institute of Information Technology **2023 – 2027**

B.Tech in Information Technology | ***CGPA: 8.97/10.00*** *Pune, India*

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, System Design, Object Oriented Programming, Machine Learning.

Experience & Leadership

India Innovates Hackathon **Jan 2026 – Feb 2026**

National Finalist & Lead Builder – Ministry of Education & AICTE *Pune, India*

- Ranked among **top 20% finalist teams** nationwide out of **6,000+ participating teams** across India, pitching scalable public service automation.
- Spearheaded product discovery and rapid prototyping, coordinating cross-functional engineering execution under tight deadline constraints.

South African Patent Office **2025**

Patent Applicant & Inventor – AI-Powered Lost & Found Matching System *Patent Filed*

- Conceptualized and filed patent for an AI matching system utilizing a **5-factor blind multimodal scoring engine** (image embeddings, NLP text, geolocation, timestamp) with automated urgency detection.
- Qualified for Round 2 in **GHCI 25 National GenAI Hackathon** (AnitaB.org India & Backbase | 'Unbound with GenAI'), demonstrating production feasibility.

Projects

Project Sentinel: AI Decision Memory & State-Drift Agent | *Python, PostgreSQL, FastAPI, Lovable, Notion* **Oct 2025 – Present**

- Spearheaded product discovery across engineering standups, Slack, and Jira, identifying critical context loss and finding that unprompted bot summaries cause heavy distraction by interrupting developer focus.
- Formulated the **"Silent-by-Default"** product thesis, passively ingesting audio transcripts and commit streams without channel spam while deliberately scoping out employee surveillance features in V1.
- Architected deterministic audit engine powered by **append-only PostgreSQL schemas**, surfacing discrepancies between spoken commitments and recorded codebase state strictly upon explicit user queries.
- Shipped clickable MVP on Lovable ([**Live Demo**]); authored comprehensive **Notion PRD** detailing state transition matrices, query grammar, and phased enterprise rollout.

CausalRetention: Prescriptive Uplift & Customer Retention Engine | *Python, XGBoost, Streamlit, Docker* **Aug 2025 – Oct 2025**

- Uncovered key economic friction in retention marketing where conventional churn models incur a **-13.1% net ROI** due to deadweight spend on non-responsive "Lost Causes" and churn acceleration from nudging dormant "Sleeping Dogs".
- Architected prescriptive AI decision engine using a **Two-Model (T-Learner) XGBoost pipeline** to calculate Conditional Average Treatment Effect (CATE), segmenting customer cohorts into 4 distinct behavioral quadrants.
- Engineered a constrained **Knapsack financial budget optimizer** under a fixed $15,000 constraint, mathematically allocating incentives to top incremental-value cohorts and capturing **+$112.7K+ in net preserved revenue (+752% campaign ROI)**.
- Shipped containerized **interactive Streamlit suite** with Docker and automated executive reporting ([**Live Demo & Repo**]); validated incrementality via Qini curves and cumulative uplift against randomized control baselines.

FlashDrop: Ultra-Low Latency Flash Sale System | *Java, Spring Boot, MySQL, Redis, Docker* **May 2025 – July 2025**

- Architected high-concurrency e-commerce checkout backend handling sudden flash-sale traffic surges with a **guaranteed 0% overselling anomaly rate** under extreme write contention.
- Engineered **5-minute automated inventory reservation locks** utilizing InnoDB pessimistic row-level locking (**SELECT FOR UPDATE**), eliminating race conditions.
- Integrated defensive transaction rollback triggers and automated deadweight cart reclamation to immediately restore stranded stock upon session expiry.
- Designed Redis cache-aside read architecture, reducing product catalog latency to **sub-25ms** during concurrent peak assessment traffic.

Technical Skills

Languages: Python, Java, SQL, C++, C, JavaScript

Frameworks: Spring Boot, FastAPI, Streamlit, XGBoost, Scikit-learn, React

Databases & Cloud: PostgreSQL, MySQL, Supabase, Redis, Docker, Git, Linux

Product & Tools: PRD Writing, Jobs-to-be-Done (JTBD), RICE, Product Metrics, Figma, Postman, Notion, LeetCode (400+ solved, 210+ GFG)

Certifications

- **Product Management: Product Management Masterclass (GeeksforGeeks)** — Product lifecycle, JTBD discovery, and roadmap metrics.
- **AI & Workflows:** Prompt Engineering | Advanced RAG (CampusX) | Docker for ML (CampusX) | n8n Automation (CampusX)